

Fibred models of dual-context type theory
and
internal universes in models of HoTT

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September 2025

Overview

- models of HoTT can be explored using a type-theoretic language (Coquand 2015, Orton and Pitts 2018)
 - for "the universe of uniform fibrations", this language must have a modality
 - ↳ "crisp type theory" - a "dual-context" type theory (Licata, Orton, Pitts and Spitters 2018)
- ⚠ this is a complication for relating its diagrammatic and type-theoretic versions.
(Awodey, Gambino and Hazratpour 2024)

GOAL: explicitly relate diagrammatic and type-theoretic constructions of the universe of uniform fibrations

Plan

① Context

- studying models of HoTT type-theoretically
- crisp type theory

② Modelling crisp type theory

- fibred model of dual-context type theory

③ Crisp type theory as an internal language

- crisp context extension
- crisp Π -types

④ Relating descriptions of the universe of uniform fibrations

① Context

Studying models of HoTT

- the simplicial sets model of HoTT (Voevodsky 2006) isn't constructive
- constructive models have subsequently been developed in cubical sets
 - Angiuli et al. 2017
 - Bezem, Coquand and Huber 2014
 - Cohen, Coquand, Huber and Mörtberg 2018
 - Awodey et al. 2025
- want to study their pros and cons

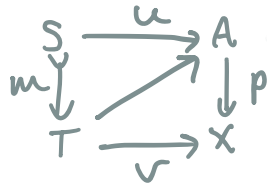
 types are modelled by fibrations with a uniformity condition
~~~~> must carry around algebraic structure

 (Coquand 2015, Orton and Pitts 2018)  
work in a type-theoretic language

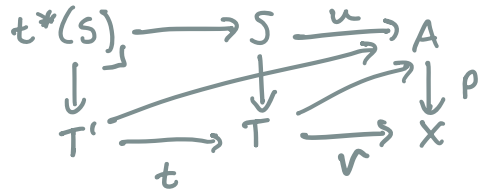
Example a "trivial fibration structure" on ...

① (category-theoretic)

$\therefore p$  is a choice of diagonal fillers  $j(m, u, v)$



for all  $m \in \text{Cof}$  such that



for all  $m \in \text{Cof}$ , for all  $t: T' \rightarrow T$ .

② (type-theoretic)

...  $\alpha: X \rightarrow U$  is an element

$t: \text{TFib}(\alpha)$

where

$$\text{TFib}(\alpha) = \prod_{q: \mathbb{Q}} \prod_{v: \alpha \{q\}} \sum_{a: \alpha} v = \lambda(a)$$

# Internal type theory

A presheaf category  $\widehat{\mathcal{C}}$

object  $\Gamma$

"small" map  $\begin{array}{c} \Gamma. \alpha \\ p \downarrow \\ \Gamma \end{array}$

Section  $\begin{array}{c} \Gamma. \alpha \\ \uparrow a \quad p \downarrow \\ \Gamma \end{array}$

extensional dependent  
type theory  $\widehat{\mathcal{L}}_{\widehat{\mathcal{C}}}$

context  $\Gamma$

type  $\Gamma \vdash \alpha \text{ type}$   
context extension  $\Gamma, x:\alpha \vdash$

term-in-context  
 $\Gamma \vdash a:\alpha$

# Relating diagrammatic and judgemental reasoning

Method 1: use the standard semantics for type theory in a locally cartesian closed category (Seely 1984, Hofmann 1994)



This is tricky for complex types like

$$\text{TFib}(\alpha) = \prod_{\varphi: \Phi} \prod_{v: \alpha^{\{\varphi\}}} \sum_{a: \alpha} v = \lambda(a)$$

Method 2: use the technique of "Kripke-Joyal forcing" (Awodey, Gambino and Hazratpour 2024)

- quasi-mechanical
- good for iterated type dependencies

## The universe of fibrations

- Recall: in cubical sets, dependent types are modelled by uniform fibrations
- type universes allow you to treat types as terms

$$A : \mathcal{U}$$

- when studying models of HoTT, not all types are fibrant
- we want a universe of fibrations / fibrant types

$$A : \mathcal{U}_{\text{Fib}}$$

to "package up" a type and its fibration structure.

! the universe of fibrations :

① in  $\hat{\mathcal{E}}$

$$\begin{array}{ccc} \text{Fib}^*(\text{id}) & \longrightarrow & \text{Fill}(\text{id} \circ -)_I \\ \downarrow \lrcorner & & \downarrow \\ \mathcal{U} & \xrightarrow{\eta} & (\mathcal{U}^I)_I \end{array}$$

② in  $\tau\hat{\mathcal{E}}$

impossible !

Problem

the internal language can't talk about part of the construction

Solution

extend the internal language with a modality  
(Licata, Orton, Pitts and Spitters (LOPS) 2018)

## Modal type theory for internal universes

(Licata, Orton, Pitts and Spitters (LOPS) 2018)

- for a type-theoretic version of the universe of fibrations, we need a distinction between "global" and "local" variables
- use a modal "dual-context" type theory

$$\Delta \mid \Gamma \vdash a : A$$

global/modal variables      local/regular variables

- global elements can be used locally, but cannot depend on local variables
- "crisp type theory", a fragment of "spatial type theory" (Shulman 2018)

## Crisp type theory

- Features "flat" modality  $\Box \alpha$



- $\Delta / \Gamma = x_1 :: \alpha_1, \dots, x_n :: \alpha_n \mid y_1 : \beta_1, \dots, y_m : \beta_m$   
(abbreviates  $x_1 : \Box \alpha_1, \dots, x_n : \Box \alpha_n, y_1 : \beta_1, \dots, y_m : \beta_m$ )
- Crisp types depend only on crisp variables
- Two kinds of context extension

① standard context extension

$$\frac{\Delta / \Gamma \vdash \alpha \text{ type}}{\Delta / \Gamma, x : \alpha \vdash}$$

② extension of the crisp context

$$\frac{\Delta / \bullet \vdash \alpha \text{ type}}{\Delta, x :: \alpha \mid \bullet \vdash}$$

+ weakening

$$\frac{\Delta / \bullet \vdash \alpha \text{ type} \quad \Delta / \Gamma \vdash \beta \text{ type}}{\Delta, x :: \alpha \mid \Gamma \vdash \beta \text{ type}}$$



- the universe construction uses

- $\Pi$ -types with crisp domain, i.e.

$$\frac{\Delta \mid \bullet \vdash \alpha \text{ type} \quad \Delta, x :: \alpha \mid \Gamma \vdash \beta \text{ type}}{\Delta \mid \Gamma \vdash \prod_{x :: \alpha} \beta \text{ type}}$$

but NOT

- the modal type operator, i.e.

$$\frac{\Delta \mid \bullet \vdash \alpha \text{ type}}{\Delta \mid \Gamma \vdash \Box \alpha \text{ type}}$$

## The problem

- There is a type-theoretic version of the universe of fibrations (LOPS 2018) using crisp type theory

### Recall:

Diagrammatic and type-theoretic descriptions can be related by

i) directly unfolding the interpretation of the internal judgements

OR

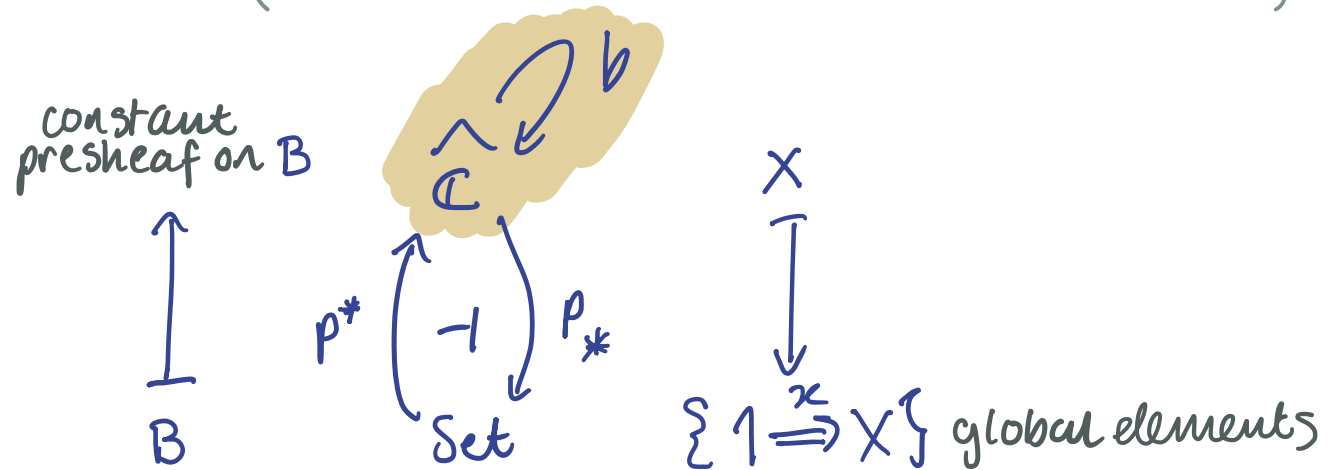
ii) quasi-mechanically, via the technique of "Kripke-Joyal forcing"



we don't have either of these for crisp type theory

# Modelling crisp type theory

Intended models (Licata et al. 2018, from Shulman 2018)



- $b$  is an idempotent comonad

# Internal crisp type theory?

A presheaf category  $\hat{\mathcal{C}}$   
idempotent comonad  $b$

?

?

?

ingredients of  
crisp type theory



dual-context  $\Delta | \Gamma$



type  $\Delta | \Gamma \vdash \alpha$  type  
context extension  $\Delta | \Gamma, x:\alpha \vdash$



term-in-context  
 $\Delta | \Gamma \vdash a:\alpha$

+ two kinds of context extension...

**GOAL:** explicitly relate diagrammatic and type-theoretic constructions of the universe of uniform fibrations in cubical set models of HoTT

**PREREQUISITE:** an internal version of crisp type theory

## Our approach

1) zoom out - how can we model the features of a dual-context type theory?

e.g. • context dependence of  $\Delta/\Gamma$   
• two kinds of context extension

2) zoom back in - how does  $\hat{\mathbb{C}}, b$  admit such a model?

3) use this understanding to extract an internal crisp type theory

4) carry out the universe construction in this internal language, taking its interpretation to relate it to the diagrammatic version.

② Modelling crisp type theory

# Modelling standard dependent type theory

(Awodey 2016, Fiore)

DEFINITION // A natural model is a category  $\mathcal{C}$  with data

i) a terminal object  $1$

ii) a representable natural transformation  $ty: \tilde{U} \rightarrow U$  in  $\hat{\mathcal{C}}$

- for all  $\Gamma \in \mathcal{C}$ ,  $\alpha \in U(\Gamma)$

there is a  $\Gamma.\alpha \in \mathcal{C}$ ,  $p: \Gamma.\alpha \rightarrow \Gamma$  and  $q \in \tilde{U}(\Gamma.\alpha)$  s.t.

$$\begin{array}{ccc} \Gamma.\alpha & \xrightarrow{q} & \tilde{U} \\ p \downarrow \lrcorner & & \downarrow ty \\ \Gamma & \xrightarrow{\alpha} & U \end{array}$$

universe of terms

universe of types

is a pullback in  $\hat{\mathcal{C}}$ .

"display map /  
representative  
for  $\alpha$ "

$$\begin{array}{c} \Gamma.\alpha \\ \downarrow p \\ \Gamma \end{array}$$

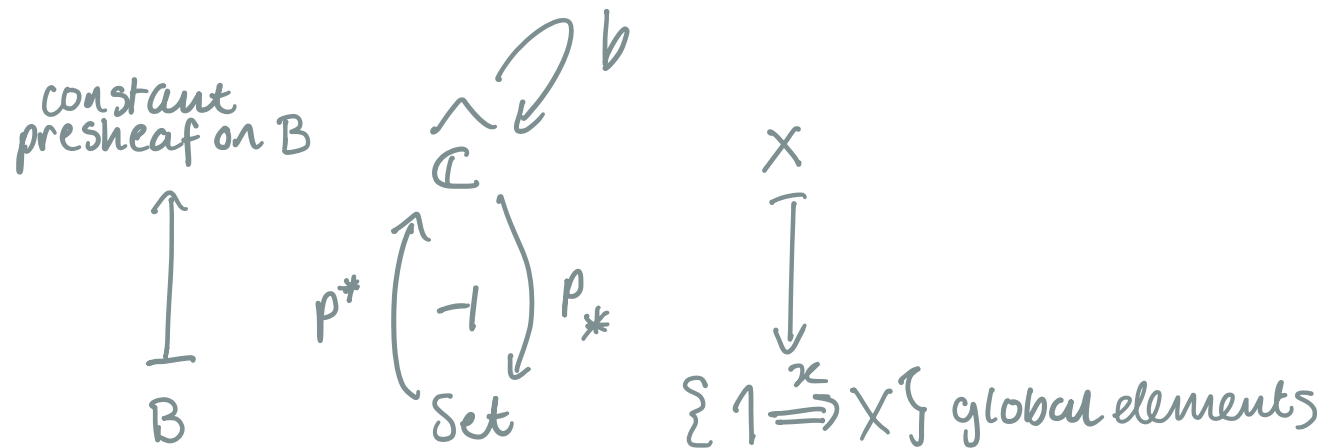


$$\begin{array}{l} \Gamma \vdash \alpha \text{ type,} \\ \Gamma.\alpha \vdash \end{array}$$



## Modelling crisp type theory

- Intended model in Licata et.al. (2018), from Shulman (2018)



- $b$  is an idempotent comonad

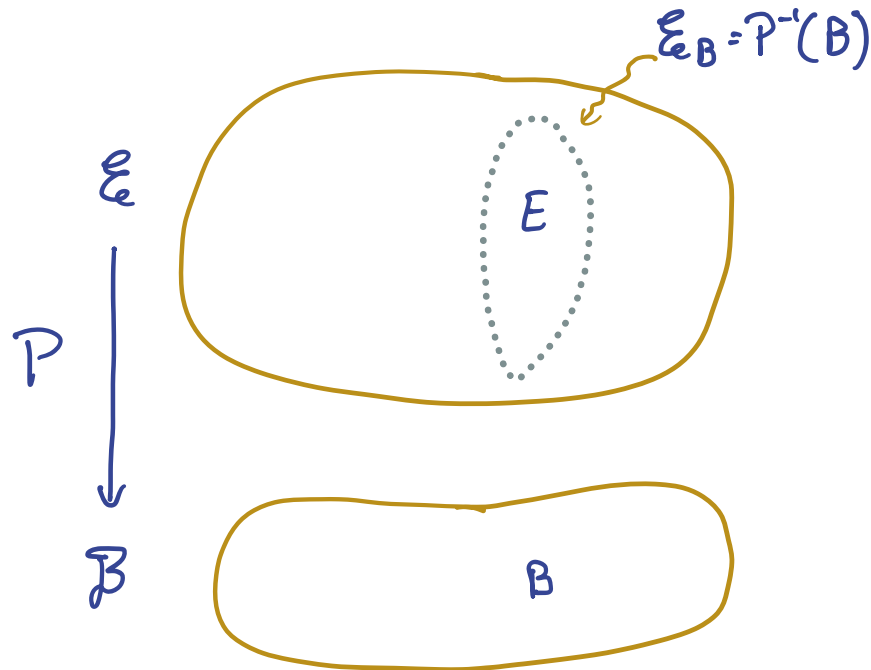
**Zooming out:** what are the features of the language and how might we model them more abstractly?

## Feature 1: Dual context

Grothendieck  
fibration

For a context  $\Delta/\Gamma$ , want to capture the dependency of  $\Gamma$  on  $\Delta$ .

Ask for a functor, viewed as a display family of categories



$$E \in \mathcal{E}_B \rightsquigarrow \Delta/\Gamma$$

$$B \in \mathcal{B} \rightsquigarrow \Delta$$

## Modelling dual-context type theory

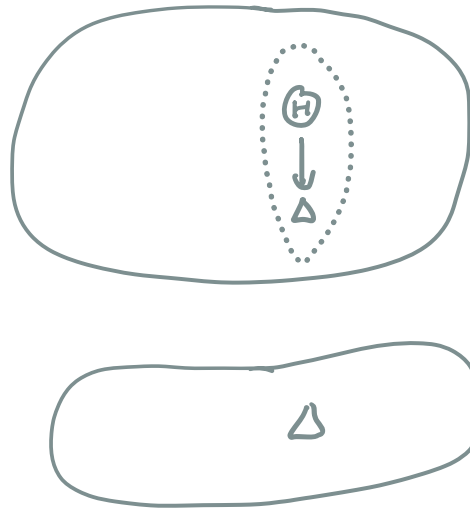


IDEA// Equip

- (i) the base category, and
- (ii) each fibre

with the structure to model a type theory.

PREVIEW// Given  $(\hat{\mathcal{C}}, b)$ , we will take -



$$\mathcal{E} := \hat{\mathcal{C}} \downarrow \hat{\mathcal{C}} b$$

$$\downarrow P := \text{cod}$$

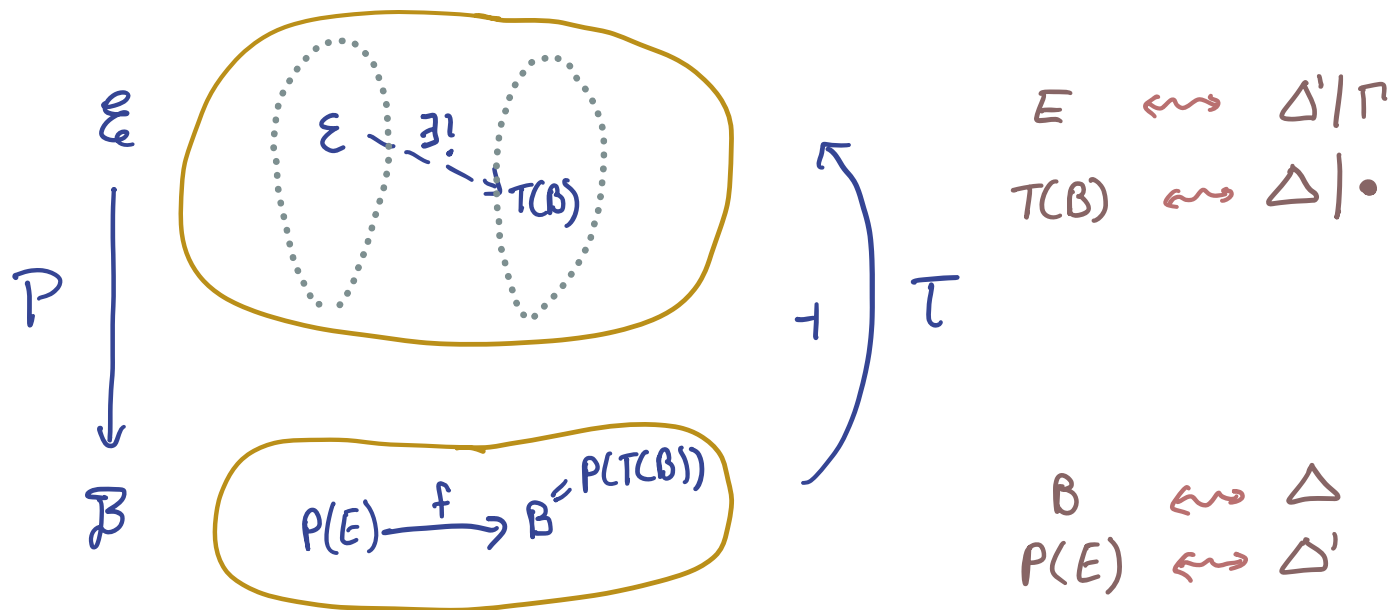
$$\mathcal{B} := \hat{\mathcal{C}} b$$

## Feature 2: Empty contexts

The second context zone may be empty:

$\Delta| \cdot$

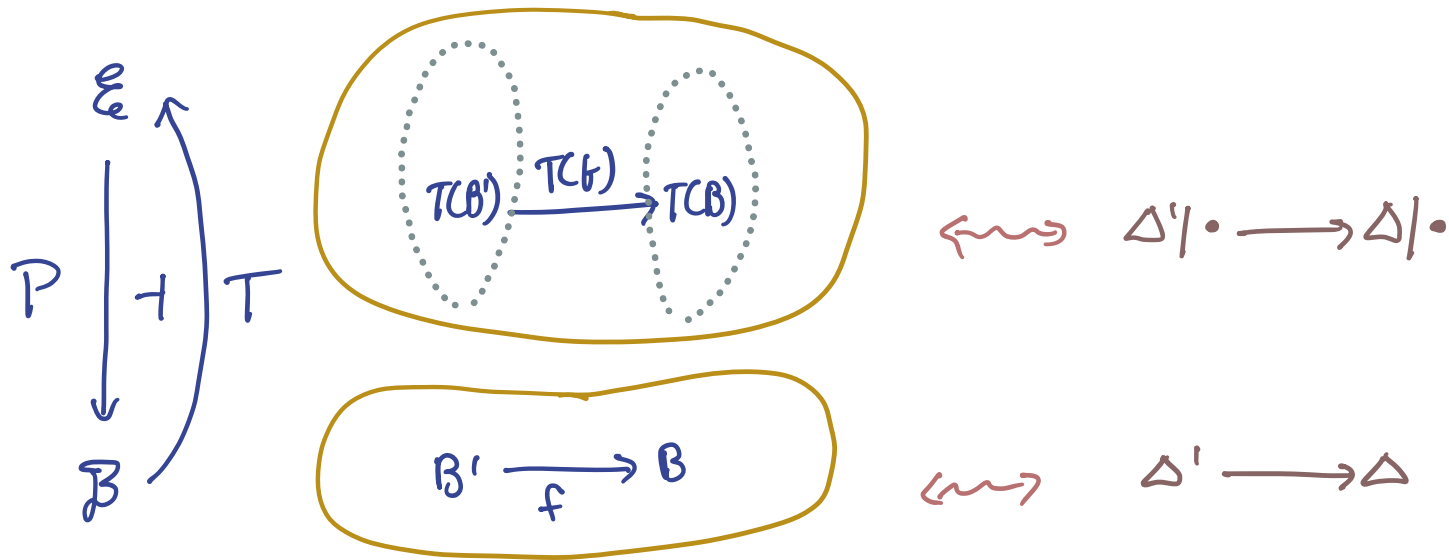
Ask for a right adjoint right inverse to  $P$ .



$\Rightarrow$  Every fibre has a terminal object

$\Rightarrow$  fibrewise terminal objects are stable under reindexing

i.e.  $T(f)$  is a cartesian lift of  $f$



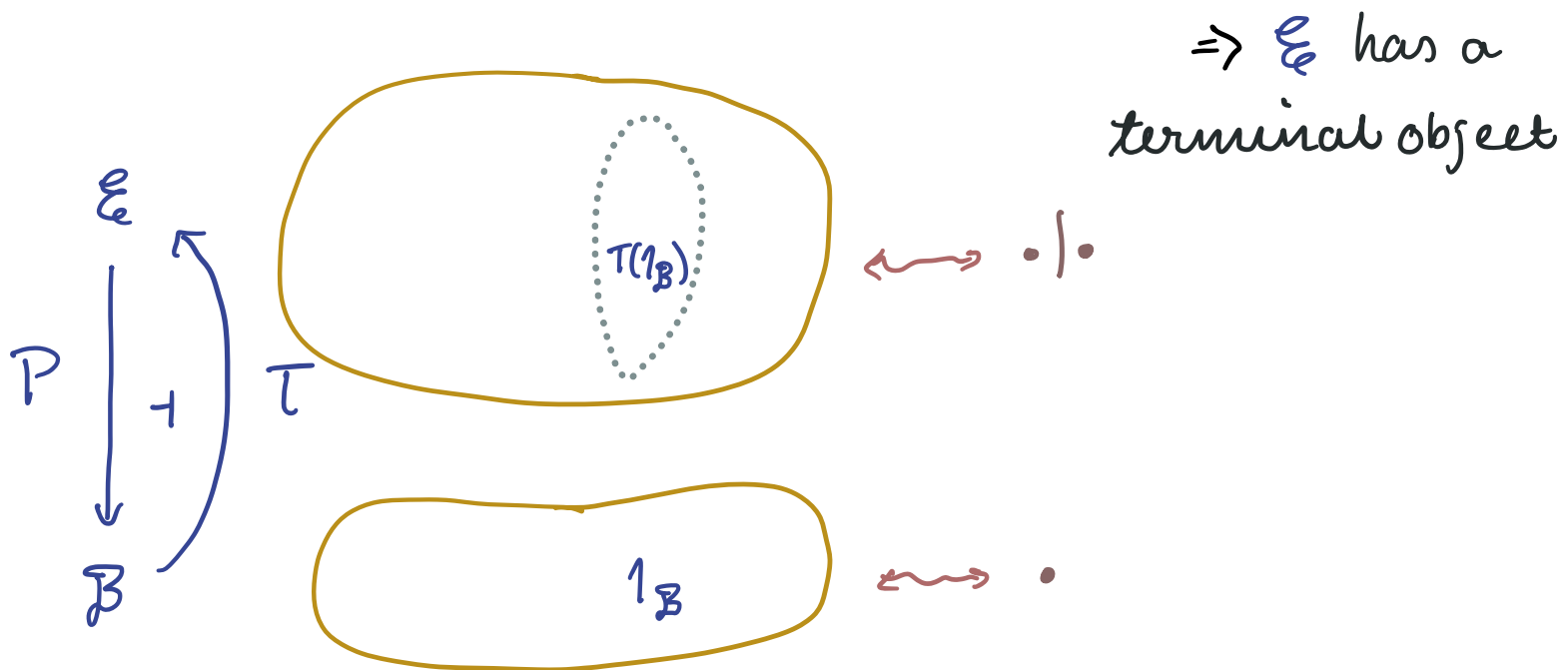
## Feature 2: Empty contexts (continued)

The context may be empty:

•/•

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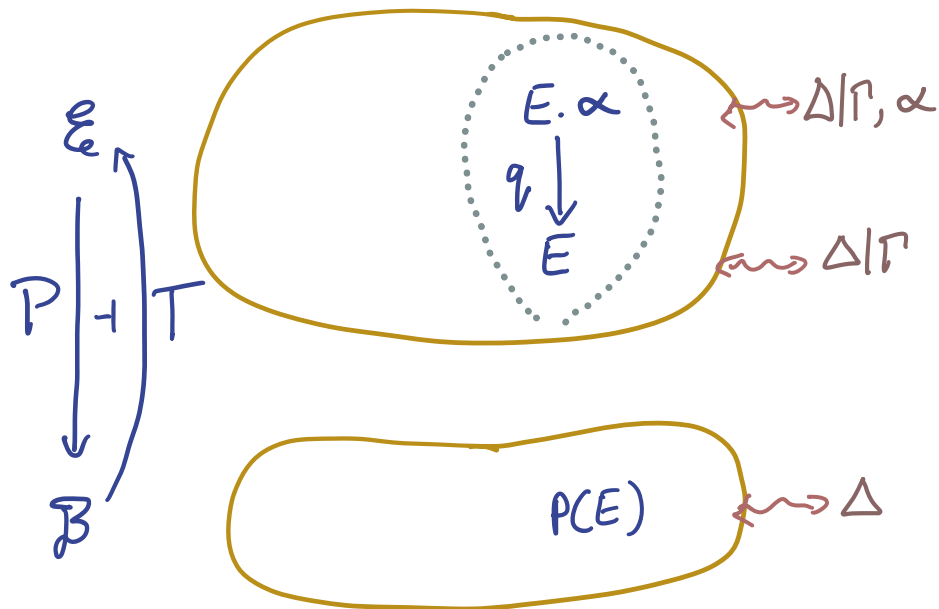
Ask that  $\mathcal{B}$  has a terminal object



## Feature 3: Extension of the second context zone

$$\frac{\Delta/\Gamma \vdash \alpha \text{ type}}{\Delta/\Gamma, \alpha \text{ context}}$$

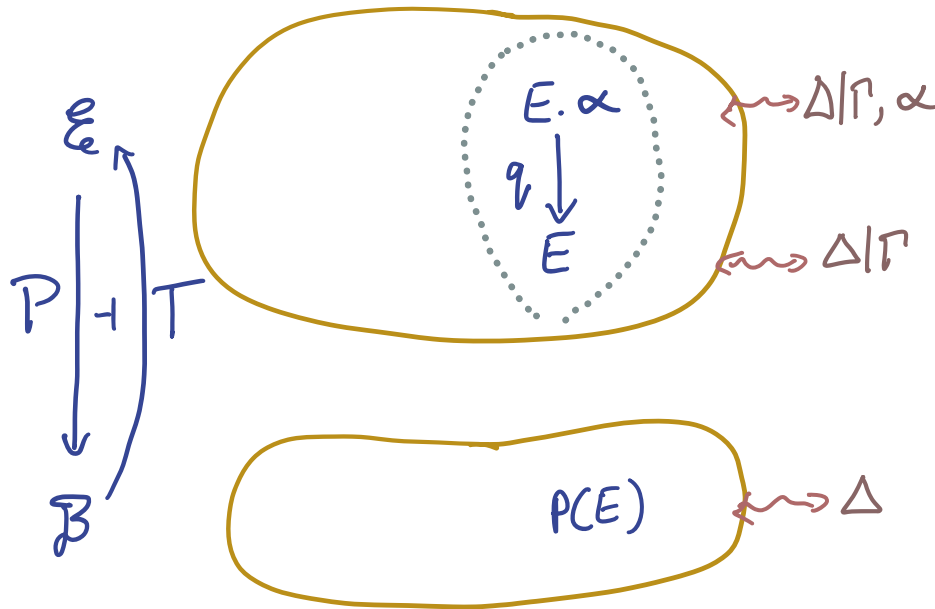
... type extension in  $\mathbb{E}$



## Feature 3: Extension of the second context zone

$$\frac{\Delta|\Gamma \vdash \alpha \text{ type}}{\Delta|\Gamma, \alpha \text{ context}}$$

... type extension in  $\mathbb{E}$



To implement:

Ask for a representable natural transformation in  $\widehat{\mathbb{E}}$  s.t. in the specified pullback along  $\alpha$ :  $E \rightarrow \mathcal{U}_{\mathbb{E}}$ ,

$$\begin{array}{ccc} E.\alpha & \xrightarrow{\quad} & \widetilde{\mathcal{U}}_{\mathbb{E}} \\ q \downarrow & \lrcorner & \downarrow \text{ty} \\ E & \xrightarrow{\alpha} & \mathcal{U}_{\mathbb{E}} \end{array}$$

$E.\alpha \xrightarrow{p} E$  in  $\mathbb{E}$  lies in the fibre  $\mathbb{E}_{P(E)}$ .



$\Rightarrow$  Each fibre has a natural model:

In the fibre  $\mathcal{E}_B$  over  $B$  there is

- a terminal object  $T(B)$
- a representable natural transformation

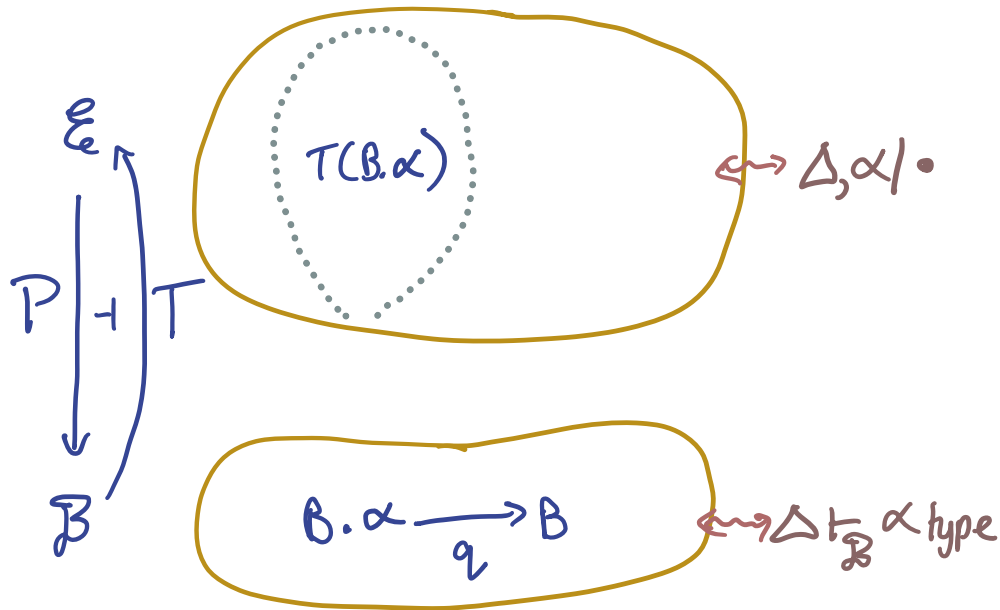
$$\tilde{u}|_{\mathcal{E}_B} \xrightarrow{t_B} u|_{\mathcal{E}_B}$$

$\Rightarrow$  The natural model structure is preserved between the fibres

## Feature 4: extension of the first context zone

$$\frac{\Delta \mid \bullet \vdash \alpha \text{ type}}{\Delta, \alpha \mid \bullet \text{ context}}$$

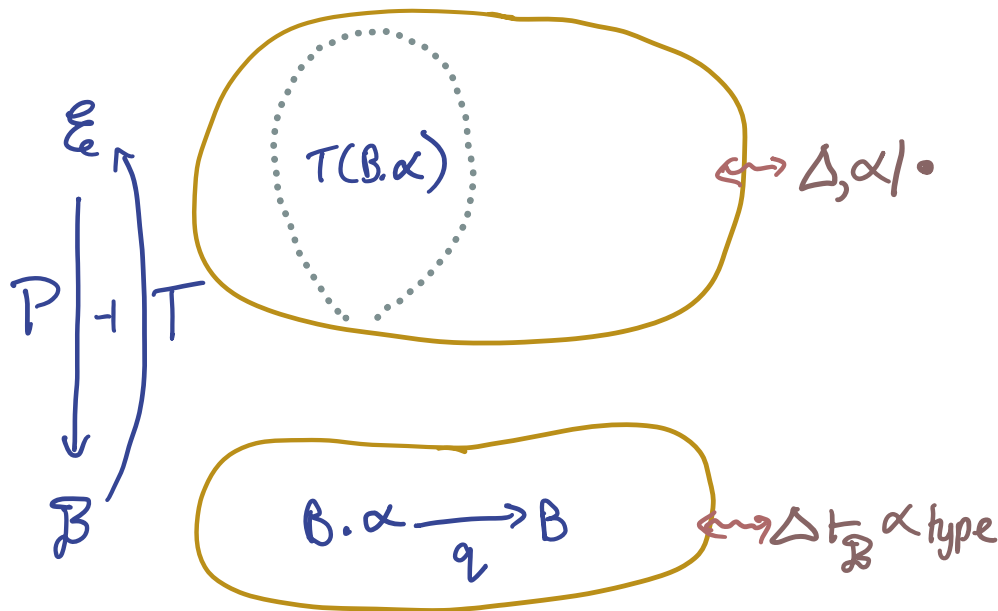
... type extension in  $\mathcal{B}$



## Feature 4: extension of the first context zone

$$\frac{\Delta | \bullet \vdash \alpha \text{ type}}{\Delta, \alpha | \bullet \text{ context}}$$

... type extension in  $\mathcal{B}$



To implement:

Ask that the map in  $\hat{\mathcal{B}}$  defined

$$\tilde{u}_B : \mathcal{B}^{op} \xrightarrow{T^{op}} \mathcal{E}^{op} \xrightarrow{\tilde{u}_\mathcal{E}} \text{Set}$$

$$\downarrow$$

$$u_B : \mathcal{B}^{op} \xrightarrow{T^{op}} \mathcal{E}^{op} \xrightarrow{u_\mathcal{E}} \text{Set}$$

is a representable natural transformation

So we have:

$$\begin{array}{ccc} B.\alpha & \xrightarrow{\quad} & \tilde{u}_\mathcal{E} \circ T^{op} \\ q \downarrow \lrcorner & & \downarrow \tau \circ T^{op} \\ B & \xrightarrow{\alpha} & u_\mathcal{E} \circ T^{op} \end{array}$$

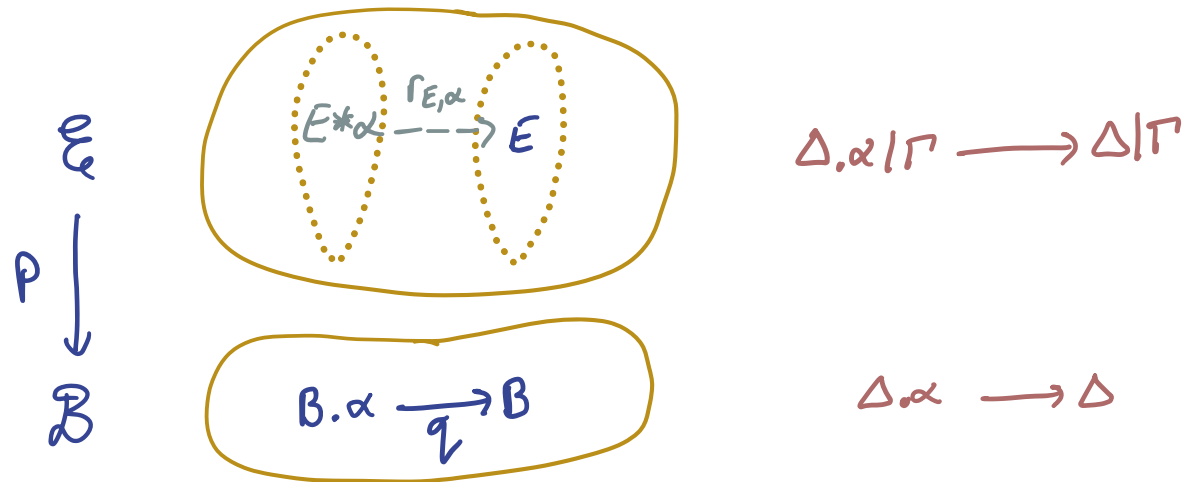
$\Rightarrow$  there is a correspondence between typing judgements  
 $\Delta \vdash_B \alpha \text{ type}$  and  $\Delta \mid \bullet \vdash_{\mathcal{E}} \alpha \text{ type}$

$$\begin{array}{lcl}
 \text{in } \hat{B}, & \frac{B \xrightarrow{\alpha} \mathcal{U}_{\mathcal{E}} \circ T^{\text{op}}}{\alpha \in \mathcal{U}_{\mathcal{E}}(T(B))} & \iff \Delta \vdash_B \alpha \text{ type} \\
 \text{in } \hat{\mathcal{E}}, & \frac{T(B) \xrightarrow{\alpha} \mathcal{U}_{\mathcal{E}}}{\alpha \in \mathcal{U}_{\mathcal{E}}(T(B))} & \iff \Delta \mid \bullet \vdash_{\mathcal{E}} \alpha \text{ type}
 \end{array}$$

## Feature 5: Weakening of the first zone

$$\frac{\Delta/\Gamma \vdash \beta \text{ type} \quad \Delta/\bullet \vdash \alpha \text{ type} \quad r: \Delta.\alpha/\Gamma \longrightarrow \Delta/\Gamma}{\Delta.\alpha/\Gamma \vdash \beta[r] \text{ type}}$$

Ask for cartesian lifts of display maps in  $\mathcal{B}$ :



Then the rule is validated by the composition

$$E * \alpha \xrightarrow{r_{E, \alpha}} E \xrightarrow{\beta} \mathcal{U} \quad \iff \quad \Delta.\alpha/\Gamma \vdash \beta[r] \text{ type}$$

# Summary

let  $P: \mathcal{E} \rightarrow \mathcal{B}$  be a functor.

## Axioms

- 1)  $P$  has a right adjoint right inverse,  $T$ .
- 2)  $\mathcal{B}$  has a terminal object.
- 3) There is a representable natural transformation  
$$\eta: \tilde{U}_{\mathcal{E}} \rightarrow U_{\mathcal{E}} \text{ in } \hat{\mathcal{E}}$$
with fibrewise display maps.
- 4)  $\tilde{U}_{\mathcal{E}} \circ T^{op} \rightarrow U_{\mathcal{E}} \circ T^{op}$  in  $\hat{\mathcal{B}}$  is representable.
- 5) There are cartesian lifts of display maps in  $\mathcal{B}$

This is used to define "fibred natural model for dual-context type theory"

## Remarks

- Axiom 5 is milder than asking for cartesian lifts of all maps in  $\mathcal{B}$  (i.e. that  $p: \mathcal{E} \rightarrow \mathcal{B}$  is a Grothendieck fibration)
  - ↳ for the term model, however, this will be true
- (Gratzer 2023) presents a similar model

## Zooming back in

THEOREM //  $\hat{\mathcal{C}}, b$  admits a fibred natural model of dual-context type theory

- Let  $\hat{\mathcal{C}}_b$  be the full subcategory of  $\hat{\mathcal{C}}$  spanned by  $M \in \text{obj}(\hat{\mathcal{C}})$  with  $bM = M$

- Let  $\hat{\mathcal{C}} \downarrow \hat{\mathcal{C}}_b$  be the comma category with objects  $\begin{array}{c} X \\ \downarrow \\ M \end{array}^x$   $\begin{array}{l} (X \in \hat{\mathcal{C}}) \\ (M \in \hat{\mathcal{C}}_b) \end{array}$

Then :

$$\begin{array}{c} X \\ \downarrow \\ M \\ \vdots \\ M \end{array} \quad \begin{array}{c} \hat{\mathcal{C}} \downarrow \hat{\mathcal{C}}_b \\ \omega d \downarrow \\ \hat{\mathcal{C}}_b \end{array} \quad + \quad \begin{array}{c} M \\ \parallel \\ M \\ \vdots \\ M \end{array} \quad \begin{array}{c} \tau \\ \uparrow \\ \downarrow \end{array} \quad + \quad \begin{array}{c} X \\ \downarrow \\ M \\ \vdots \\ bX \end{array} \quad \begin{array}{c} S \\ \downarrow \end{array}$$



③ Crisp type theory as an  
internal language

## Approach

- (0) Understand how  $\hat{\mathcal{C}} \downarrow \hat{\mathcal{C}}_b$  models a dual-context type theory
- (1) Define an internal type theory for  $\hat{\mathcal{C}} \downarrow \hat{\mathcal{C}}_b$ , by analogy with the usual presheaf case
- (2) Show that the rules of crisp type theory are valid in  $\hat{\mathcal{C}} \downarrow \hat{\mathcal{C}}_b$

Focus on:

- crisp context extension
- crisp  $\Pi$ -types

## The internal type theory $\mathcal{T}\hat{\mathcal{C}}$

Let  $\hat{\mathcal{C}}$  be a presheaf category,  $\mathcal{S}$  the class of "small maps"

### PRELIMINARIES//

$\hat{\mathcal{C}}$  has a small map classifier  $\pi: E \longrightarrow U$  (Hofmann & Streicher 1997)

- for every  $p: T' \rightarrow T$  in  $\mathcal{S}$   
there is a classifying map  $\alpha: T \rightarrow U$

$$\begin{array}{ccc} T' & \xrightarrow{\quad} & E \\ \downarrow p & \lrcorner & \downarrow \pi \\ T & \xrightarrow{\alpha} & U \end{array}$$

DEFINITION //  $(\mathcal{T}\hat{\mathcal{C}})$

- a CONTEXT is an object  $\Gamma$  of  $\hat{\mathcal{C}}$
- a TYPE  $\alpha$  in context  $\Gamma$  is a map  $\Gamma \xrightarrow{\alpha} U$
- a TERM  $a$  of type  $\alpha$  in context  $\Gamma$  is a map

$$a: \Gamma \longrightarrow E$$

such that

$$\begin{array}{ccc} & a \nearrow & E \\ \Gamma & \xrightarrow{\alpha} & U \\ & \searrow \pi & \end{array}$$

DEFINITION// ( $\mathcal{T}\hat{\mathcal{C}}$ )

- a CONTEXT is an object  $\Gamma$  of  $\hat{\mathcal{C}}$
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such that

$$\begin{array}{ccc} & a \nearrow & E \\ \Gamma & \xrightarrow{\alpha} & \mathcal{U} \\ & \searrow \pi & \end{array}$$

NOTATION//

$$\Gamma \vdash$$

$$\Gamma \vdash \alpha: \mathcal{U}$$

$$\Gamma \vdash a: \alpha$$

## CONTEXT EXTENSION //

If  $\alpha$  is a type in context  $\Gamma$ ,  
then  $\Gamma.\alpha$  is a context

$$\text{NOTATION // } \frac{\Gamma \vdash \alpha : U}{\Gamma.\alpha \vdash}$$

$\langle \Rightarrow \rangle$

$$\begin{array}{ccc} \Gamma.\alpha & \longrightarrow & E \\ \downarrow \perp & & \downarrow \pi \\ \Gamma & \xrightarrow{\alpha} & U \end{array}$$

## SUBSTITUTION //

$$\frac{t : \Delta \rightarrow \Gamma \quad \Gamma \vdash \alpha : U}{\Delta \vdash \alpha(t) : U}$$

$\langle \Rightarrow \rangle$

$$\Delta \xrightarrow{t} \Gamma \xrightarrow{\alpha} U$$

$$\frac{t : \Delta \rightarrow \Gamma \quad \Gamma \vdash a : \alpha}{\Delta \vdash a(t) : \alpha(t)}$$

$\langle \Rightarrow \rangle$

$$\begin{array}{ccc} & & E \\ & \nearrow & \downarrow \pi \\ \Delta \xrightarrow{t} \Gamma & \xrightarrow{\alpha} & U \end{array}$$

## $\Sigma$ and $\Pi$ types in $\hat{\mathcal{C}}$

FACT//  $\hat{\mathcal{C}}$  validates rules corresponding to  $\Sigma$  and  $\Pi$  types

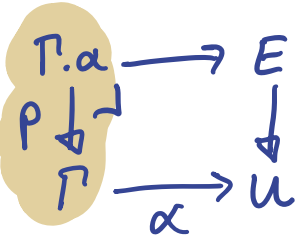
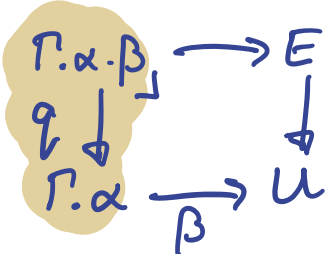
e.g. 
$$\frac{\Gamma \vdash \alpha : U \quad \Gamma, \alpha \vdash \beta : U}{\Gamma \vdash \prod_{\alpha} \beta : U} \text{ is valid in } \hat{\mathcal{C}}$$

IDEA// -  $\hat{\mathcal{C}}$  is locally cartesian closed

- left and right adjoints to pullback along any map restrict to small maps

want 
$$\frac{\Gamma \vdash \alpha : U \quad \Gamma.\alpha \vdash \beta : U}{\Gamma \vdash \Pi_{\alpha} \beta : U}$$

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①  $\Gamma \vdash \alpha : U \iff$    $\Gamma.\alpha \vdash \beta : U \iff$  

② we have

$$\begin{array}{ccc} \mathcal{S}/\Gamma & \xrightarrow{\rho^*} & \mathcal{S}/\Gamma.\alpha \\ & \downarrow \pi & \\ \Pi_{\rho}(\Gamma.\alpha.\beta) & & \Gamma.\alpha.\beta \\ \Pi_{\rho} q \downarrow & \Gamma \dashv \vdash & \downarrow q \\ \Gamma & & \Gamma.\alpha \end{array}$$

③  $\Pi_{\rho} q$  is small, so is classified by a map

$$\Gamma \xrightarrow{\Pi_{\alpha} \beta} U \iff \Gamma \vdash \Pi_{\alpha} \beta : U$$

④ + introduction, elimination and computation rules



## The internal type theory $\mathcal{T} \in \mathcal{L} \mathcal{E} \mathcal{B}$

Let  $\mathcal{E}$  be a presheaf category,  $\mathcal{S}$  the class of "small maps"

Recall:  $\mathcal{E} \downarrow \mathcal{E}_b$  has objects  $\begin{array}{c} \textcircled{H} \\ \downarrow x \\ \Delta \end{array}$  with  $b\Delta = \Delta$

### PRELIMINARIES

DEFINITION // a **VERTICAL SMALL MAP** is a map of the form

$$\begin{array}{ccc} & \textcircled{H'} & \\ p \swarrow & \downarrow x' & \\ \textcircled{H} & & \Delta \\ x \downarrow & \searrow & \\ \Delta & & \end{array} \quad \text{where } p: \textcircled{H'} \rightarrow \textcircled{H} \in \mathcal{S}$$

FACT // There is a classifier of vertical small maps

$$\begin{array}{ccccc} & & \textcircled{H'} & \xrightarrow{\quad} & E \\ & \swarrow & \downarrow & \nearrow \pi & \downarrow \\ \textcircled{H} & \xrightarrow{\alpha} & U & \xrightarrow{\quad} & 1 \\ \downarrow & \searrow & \downarrow & \nearrow & \downarrow \\ \Delta & \xrightarrow{\quad} & 1 & \xrightarrow{\quad} & 1 \end{array}$$

# DEFINITION //

• a CONTEXT is an object  $\begin{array}{c} \textcircled{H} \\ \downarrow \\ \Delta \end{array} \chi$  of  $\hat{\mathcal{C}}\hat{\mathcal{L}}\hat{\mathcal{C}}_b$ .

• a TYPE  $\alpha$  in context  $\chi$  is a map  $(\alpha, !_\Delta)$

$$\begin{array}{ccc} \textcircled{H} & \xrightarrow{\alpha} & U \\ \chi \downarrow & & \downarrow \\ \Delta & \longrightarrow & 1 \end{array}$$

• a TERM  $a$  of type  $\alpha$  in context  $\Gamma$  is map  $(a, !_\Delta)$

$$\begin{array}{ccccc} & & \alpha & \longrightarrow & E \\ & \nearrow & & & \downarrow \\ \textcircled{H} & \xrightarrow{\alpha} & U & & 1 \\ \chi \downarrow & & \downarrow & \nearrow & \parallel \\ \Delta & \longrightarrow & 1 & & \end{array}$$

## DEFINITION //

• a CONTEXT is an object  $\begin{array}{c} \textcircled{H} \\ \downarrow \\ \Delta \end{array} \chi$  of  $\hat{\mathcal{C}}\hat{\mathcal{L}}\hat{\mathcal{C}}_b$ .

• a TYPE  $\alpha$  in context  $\chi$  is a map  $(\alpha, !_\Delta)$

$$\begin{array}{ccc} \textcircled{H} & \xrightarrow{\alpha} & U \\ \chi \downarrow & & \downarrow \\ \Delta & \longrightarrow & 1 \end{array}$$

• a TERM  $a$  of type  $\alpha$  in context  $\Gamma$  is map  $(a, !_\Delta)$

$$\begin{array}{ccccc} & & \alpha & \rightarrow & E \\ & \nearrow & & & \downarrow \\ \textcircled{H} & \xrightarrow{\alpha} & U & & 1 \\ \chi \downarrow & & \downarrow & \nearrow & \parallel \\ \Delta & \longrightarrow & 1 & & \end{array}$$

## NOTATION //

$$\textcircled{H} \vdash_{\Delta}$$

$$\textcircled{H} \vdash_{\Delta} \alpha \text{ type}$$

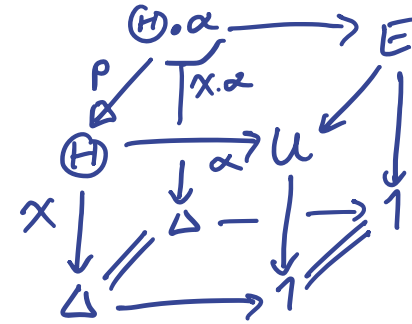
$$\textcircled{H} \vdash_{\Delta} a : \alpha$$

## STANDARD CONTEXT EXTENSION //

If  $\alpha$  is a type in context  $\mathcal{X}$ ,  
then  $\mathcal{X}.\alpha$  is a context

NOTATION // 
$$\frac{\Theta \vdash_{\Delta} \alpha \text{ type}}{\Theta.\alpha \vdash_{\Delta}}$$

$\Leftrightarrow$



$(p, id_{\Delta})$  is a vertical small map

## CRISP CONTEXT EXTENSION //

If  $\alpha$  is a crisp type, i.e. a type

in context  $\begin{array}{c} \Delta \\ \parallel \\ \Delta \end{array} id_{\Delta}$ , we have:

$\Leftrightarrow$



$$(i) \quad \frac{\Delta \vdash_{\Delta} \alpha \text{ type}}{\Delta.\alpha \vdash_{\Delta} \alpha}$$

$$(ii) \quad \frac{\Delta \vdash_{\Delta} \alpha \text{ type} \quad \Theta \vdash_{\Delta}}{\Theta * \alpha \vdash_{\Delta} \alpha}$$

want (i)  $\frac{\Delta \vdash_{\Delta} \alpha \text{ type}}{\Delta.\alpha \vdash_{\Delta.\alpha}}$  and (ii)  $\frac{\Delta \vdash_{\Delta} \alpha \text{ type} \quad \textcircled{H} \vdash_{\Delta}}{\textcircled{H} * \alpha \vdash_{\Delta.\alpha}}$

---

①  $\Delta \vdash_{\Delta} \alpha \text{ type} \quad \Leftrightarrow$

$$\begin{array}{ccc} \Delta & \xrightarrow{\alpha} & u \\ \parallel & & \downarrow \\ \Delta & \longrightarrow & 1 \end{array}$$

② apply  $s: \hat{\mathcal{C}} \downarrow \hat{\mathcal{C}}_b \rightarrow \hat{\mathcal{C}}_b$ :

$$\Delta = b\Delta \xrightarrow{b\alpha} bu$$

③ there is a classifier of small maps in  $\hat{\mathcal{C}}_b$ ; pullback along  $b\alpha$ :

$$\begin{array}{ccc} \Delta.b\alpha & \longrightarrow & bE \\ \downarrow q_{b\alpha} \quad \lrcorner & & \downarrow \\ \Delta & \xrightarrow{b\alpha} & bu \end{array}$$

④ take the *wcd*-cartesian lift of  $q_{b\alpha}$  to any context with crisp component  $\Delta$

want (i)  $\frac{\Delta \vdash_{\Delta} \alpha \text{ type}}{\Delta \cdot \alpha \vdash_{\Delta \cdot \alpha}}$  and (ii)  $\frac{\Delta \vdash_{\Delta} \alpha \text{ type} \quad \textcircled{H} \vdash_{\Delta}}{\textcircled{H} * \alpha \vdash_{\Delta \cdot \alpha}}$

---

④ take the *wd*-cartesian lift of  $q_{b\alpha}$  to any context with crisp component  $\Delta$

$$\begin{array}{c} \hat{\mathbb{C}} \downarrow \hat{\mathbb{C}}_b \\ \text{wd} \downarrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow \\ \hat{\mathbb{C}}_b \quad \quad \quad \tau \quad \quad \quad s \end{array}$$

Recall (from the abstract model): display maps in the base have cartesian lifts

- here, this is given by pullback

$$\begin{array}{ccc} \textcircled{H} * \alpha & \xrightarrow{\quad} & \textcircled{H} \\ \downarrow \lrcorner & & \downarrow \chi \\ \Delta \cdot \alpha & \xrightarrow{q_{b\alpha}} & \Delta \end{array}$$

Want (i)  $\frac{\Delta \vdash_{\Delta} \alpha \text{ type}}{\Delta.\alpha \vdash_{\Delta.\alpha}}$  and (ii)  $\frac{\Delta \vdash_{\Delta} \alpha \text{ type} \quad \textcircled{H} \vdash_{\Delta}}{\textcircled{H} * \alpha \vdash_{\Delta.\alpha}}$

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SUMMARY//

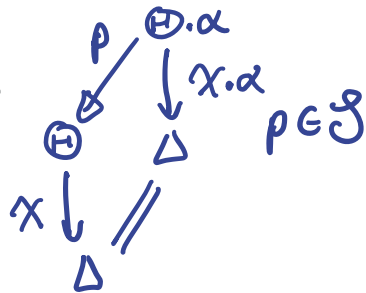
$\Delta \vdash_{\Delta} \alpha \text{ type} \quad \Leftrightarrow \quad \Delta.\alpha \xrightarrow{q_{\alpha}} \Delta \text{ in } \hat{\mathcal{C}}_b$

$\Delta.\alpha \vdash_{\Delta.\alpha} \quad \Leftrightarrow \quad \begin{array}{ccc} \Delta.\alpha & \rightarrow & \Delta \\ \parallel \lrcorner & & \parallel \\ \Delta.\alpha & \xrightarrow{q_{\alpha}} & \Delta \end{array} \text{ in } \hat{\mathcal{C}} \downarrow \hat{\mathcal{C}}_b$

$\textcircled{H} * \alpha \vdash_{\Delta.\alpha} \quad \Leftrightarrow \quad \begin{array}{ccc} \textcircled{H} * \alpha & \rightarrow & \textcircled{H} \\ \chi * \alpha \downarrow \lrcorner & & \downarrow \chi \\ \Delta.\alpha & \xrightarrow{q_{\alpha}} & \Delta \end{array} \text{ in } \hat{\mathcal{C}} \downarrow \hat{\mathcal{C}}_b$

## Vertical and horizontal small maps

vertical  
small maps

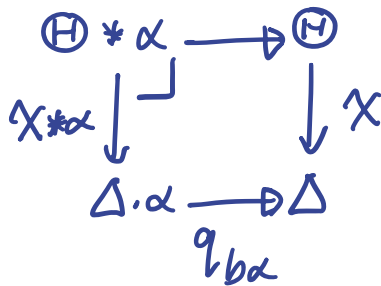


$\Leftrightarrow$

standard context extension

$$\frac{\Delta | \Gamma \vdash \alpha \text{ type}}{\Delta | \Gamma, x : \alpha \vdash}$$

horizontal small maps



$\Leftrightarrow$

crisp context extension

$$\frac{\Delta | \bullet \vdash \alpha \text{ type} \quad \Delta | \Gamma \vdash}{\Delta, x :: \alpha | \Gamma \vdash}$$



## The internal type theory $\mathcal{T}_{\mathcal{E}\downarrow\mathcal{E}_b}$

we have seen

- basic definition
- standard and crisp context extension are valid in  $\mathcal{E}\downarrow\mathcal{E}_b$

using this notation,  $\mathcal{E}\downarrow\mathcal{E}_b$  also validates rules for standard and crisp versions of

- substitution
- weakening

What about type constructors?

## The internal type theory $\hat{\mathcal{C}}\downarrow\hat{\mathcal{C}}_b - \Sigma$ and $\Pi$ types

THEOREM// ①  $\hat{\mathcal{C}}\downarrow\hat{\mathcal{C}}_b$  validates rules corresponding to  $\Sigma$  and  $\Pi$  types

e.g. 
$$\frac{\textcircled{H} \vdash_{\Delta} \alpha \text{ type} \quad \textcircled{H}. \alpha \vdash_{\Delta} \beta \text{ type}}{\textcircled{H} \vdash_{\Delta} \prod_{\alpha} \beta : \mathcal{U}}$$
 is valid in  $\hat{\mathcal{C}}\downarrow\hat{\mathcal{C}}_b$

②  $\hat{\mathcal{C}}\downarrow\hat{\mathcal{C}}_b$  validates rules corresponding to **CRISP  $\Pi$  types**

e.g. 
$$\frac{\Delta \vdash_{\Delta} \alpha \text{ type} \quad \textcircled{H} \vdash_{\Delta.\alpha} \beta \text{ type}}{\textcircled{H} \vdash_{\Delta} \prod_{\alpha}^{\text{crisp}} \beta \text{ type}}$$
 is valid in  $\hat{\mathcal{C}}\downarrow\hat{\mathcal{C}}_b$

IDEA// -  $\hat{\mathcal{C}}\downarrow\hat{\mathcal{C}}_b$  is locally cartesian closed

- left and right adjoints to pullback along any map  $(p, q)$   
in  $\hat{\mathcal{C}}\downarrow\hat{\mathcal{C}}_b$  restrict to **vertical small maps**

- if  $(p, q)$  is a vertical small map  $\rightsquigarrow$  standard  $\Pi$ -types
- if  $(p, q)$  is a horizontal small map  $\rightsquigarrow$  crisp  $\Pi$ -types

## Working with $\tau_{\hat{\mathcal{E}} \downarrow \hat{\mathcal{E}}_b}$

- it is easier to work with  $\tau_{\hat{\mathcal{E}}}$ , and often we can

-  $\tau_{\hat{\mathcal{E}} \downarrow \hat{\mathcal{E}}_b}$  simplifies to  $\tau_{\hat{\mathcal{E}}}$  when:

(i) the crisp context is empty,  $\textcircled{H} \vdash_1$

(Since  $\hat{\mathcal{E}} \hookrightarrow \hat{\mathcal{E}} \downarrow \hat{\mathcal{E}}_b$  is a reflective full subcategory)

(ii) the non-crisp context is empty,  $\Delta \vdash_{\Delta}$

(Since  $\hat{\mathcal{E}}_b \hookrightarrow \hat{\mathcal{E}} \downarrow \hat{\mathcal{E}}_b$  is a coreflective full subcategory)

④ Relating descriptions of the universe of uniform fibrations

## Uniform fibrations

- Introduced to give a constructive model of univalence, in cubical sets  
(Bezem, Coquand and Huber 2014 ;  
Cohen, Coquand, Huber and Mörtberg 2018)
- Dependent types are modelled by Kan fibrations with a uniformity condition
- Developments :

### Diagrammatic

- Swan 2016
- Gambino and Sattler 2017
- Awodey 2023

### Type-theoretic

- Orton and Pitts 2018
- Licata, Orton, Pitts & Spitters 2018



Related by Awodey, Gambino and Hazratpour 2024,  
EXCEPT FOR the universe of uniform fibrations

# The type of uniform fibrations

(Following AGH2024)



Package up fibration structures on a map into a type

In  $\hat{\mathcal{C}}$

$$\text{i)} \quad \begin{array}{ccc} X \cdot \alpha & \longrightarrow & E \\ p_\alpha \downarrow \lrcorner & & \downarrow \\ X & \xrightarrow{\alpha} & U \end{array}$$

In  $\hat{\mathcal{C}}$

$$\langle \rightsquigarrow \rangle \quad X \vdash \alpha : U$$

the interval

ii)

$$X^I \xrightarrow{\text{Fill}(\alpha_0)} U$$

$$\langle \rightsquigarrow \rangle \quad X^I \vdash \text{Fill}(\alpha_0) : U$$

“Filling structure”

$$\text{iii)} \quad \begin{array}{ccc} X^I \cdot \text{Fill}(\alpha_0) & \longrightarrow & E \\ f \uparrow \lrcorner & & \downarrow \\ X^I & \xrightarrow{\text{Fill}(\alpha_0)} & U \end{array}$$

$$\langle \rightsquigarrow \rangle \quad X^I \vdash f : \text{Fill}(\alpha_0)$$

$$\text{iii)} \quad \begin{array}{ccc} X^I \cdot \text{Full}(\alpha \circ -) & \longrightarrow & E \\ \downarrow & \lrcorner & \downarrow \\ X^I & \xrightarrow{\text{Full}(\alpha \circ -)} & U \end{array} \quad \leadsto \quad X^I \vdash f: \text{Fill}(\alpha \circ -)$$

$$\text{iv)} \quad \begin{array}{ccc} \prod_{x: X^I} \text{Fill}(\alpha \circ -) \\ \downarrow \\ 1 \end{array} \quad \leadsto \quad \bullet \vdash g: \prod_{x: X^I} \text{Fill}(\alpha \circ -)$$

DEF  $\text{Fib}(\alpha) := \prod_{x: X^I} \text{Fill}(\alpha \circ -)$

## Universe of uniform fibrations

(Following AGH2024)

- $\text{Fib}(\alpha)$  is indexed over  $X^I$ , not  $X$
- Need a pullback-stable family over  $X$
- use the fact that  $(-)^I$  has a right adjoint,  $\sqrt{(-)}$

"tiny interval"

$$\begin{array}{ccccc}
 \text{Fib}^*(\alpha) & \xrightarrow{\quad} & \sqrt{X^I \cdot \text{Full}(\alpha \circ -)} & \xrightarrow{\quad} & \sqrt{E} \\
 \downarrow \lrcorner & & \downarrow \lrcorner & & \downarrow \\
 X & \xrightarrow{\quad \eta \quad} & \sqrt{X^I} & \xrightarrow[\sqrt{\text{Full}(\alpha \circ -)}]{} & \sqrt{U}
 \end{array}$$

- To get the universe: let  $\alpha$  be  $\text{id}: U \rightarrow U$
- we obtain a "universe" i.e. a uniform fibration classifier



## Internalising the tiny interval adjunction

- the interval in the category  $\hat{\square}$  of cubical sets is **tiny**,  $(-)^I \dashv \sqrt{(-)}$
- an "internal" version of this adjunction replaces hom-sets with exponential objects, i.e.

instead of  $\hat{\square}(A^I, B) \cong \hat{\square}(A, \sqrt{B})$

want a family  $(-)^I \Rightarrow B \cong_{\alpha_B} (-) \Rightarrow \sqrt{B}$

$\begin{array}{c} \text{(i)} \\ \uparrow \\ \text{(ii)} \end{array} \quad \begin{array}{c} \text{X} \\ \downarrow \end{array}$

- The internal version is equivalent to having natural isomorphisms

$$\hat{\square}(k_c, A^I \Rightarrow B) \cong \hat{\square}(k_c, A \Rightarrow \sqrt{B})$$

- we only have

$$\hat{\square}(1, A^I \Rightarrow B) \cong \hat{\square}(1, A \Rightarrow \sqrt{B})$$

- Therefore, we only have an internal version of  $(-)^I \dashv \sqrt{(-)}$  when we restrict to **global elements**, i.e. crisp types

## Towards an explicitly internal universe

- (LOPS18) constructs the universe of uniform fibrations in crisp type theory
- given an internal version of crisp type theory, we can now explicitly relate this to the diagrammatic version in (AGH24)

THEOREM // The  $\hat{\mathcal{C}} \downarrow \hat{\mathcal{C}}_b$ -judgement

$$\sum_{x:X} \sum_{y: \sqrt{x^{\mathcal{F}}}. bFill(\alpha \circ x)} R(id')(x) =_{\sqrt{x^{\mathcal{F}}}} \sqrt{p}(y) \mid \bullet \vdash$$

is interpreted by the object  $Fib^*(\alpha)$  in  $\hat{\mathcal{C}} \downarrow \hat{\mathcal{C}}_b$

THEOREM // The  $\tau_{\hat{\mathcal{E}} \downarrow \hat{\mathcal{E}}_b}$ -judgement

$$\sum_{x:X} \sum_{y: \sqrt{x^I} \cdot b \text{Fill}(\alpha \circ x)} R(\text{id}') (x) =_{\sqrt{x^I}} \sqrt{\rho}(y) \mid \bullet \vdash$$

is interpreted by the object  $\text{Fib}^*(\alpha)$  in  $\hat{\mathcal{E}} \downarrow \hat{\mathcal{E}}_b$

IDEA // - show the axioms for tinyness of the interval (LOPS18) are valid in  $\hat{\mathcal{E}} \downarrow \hat{\mathcal{E}}_b$

e.g.  $\bullet \mid \bullet \vdash \prod_{f :: \alpha^I \Rightarrow \beta}^{\text{crisp}} R(f) : \alpha \Rightarrow \sqrt{\beta}$

- use a mixture of reasoning in  $\tau_{\hat{\mathcal{E}} \downarrow \hat{\mathcal{E}}_b}$  and  $\tau_{\hat{\mathcal{E}}}$

## Next step

- Recall: the (diagrammatic) universe was  $\text{Fib}^*(\text{id})$ ,

$$\begin{array}{ccccc}
 \text{Fib}^*(\text{id}) & \xrightarrow{\quad} & \sqrt{\mathcal{U}^E \cdot \text{Full}(\text{id}_E)} & \xrightarrow{\quad} & \sqrt{E} \\
 \downarrow \lrcorner & & \downarrow \lrcorner & & \downarrow \\
 \mathcal{U} & \xrightarrow{\quad \eta \quad} & \sqrt{\mathcal{U}^E} & \xrightarrow[\sqrt{\text{Full}(\text{id}_E)}]{} & \sqrt{\mathcal{U}}
 \end{array}$$

in which  $\sqrt{\quad}$  is applied to  $\mathcal{U}$ .

- For the type-theoretic version, this requires a hierarchy of type universes
- Remains to adapt  $\tau_{\mathcal{A} \downarrow \mathcal{B}}$  to include a hierarchy
- Nevertheless, the  $\text{Fib}^*(\alpha)$  result illustrates the concept.

## Summary

- developed a fibred model of dual-context type theory
- used this understanding to extract crisp type theory as an internal language of a category
- illustrated how we can now explicitly relate diagrammatic and type-theoretic descriptions of the universe of uniform fibrations
  - ↳ expands on
    - (LOPS18), via a more detailed model,
    - (AGH24), by not being restricted to working in  $\tau_E$ .

## Further work

- adapt  $\tau_{\hat{a} \downarrow \hat{b}}$  to include a hierarchy
- explore crisp Kripke-Joyal forcing
- consider the not-necessarily-crisp universe of fibrations in (Riley 2024)

Thank you!